HM Government

T-LEVELS

Student Workbook: The Haber Process

T Level Technical Qualification in Science • Chemistry • A12.6

Name: _____ Date: _____

Specification coverage

Point	Required understanding
A12.6.1	Haber process: conditions and economic effect of conditions; importance of compromising between equilibrium and reaction rate.

1. Retrieval

1.1 Equilibrium

For an exothermic forward reaction, what is the effect of lowering temperature on equilibrium position?

_____

_____

_____

1.2 Pressure

What does increasing pressure favour in a gaseous equilibrium?

_____

_____

_____

1.3 Catalyst

What does a catalyst do to activation energy, and does it change equilibrium position?

_____

_____

2. The Haber process

2.1 Equation

Write the balanced reversible equation for the Haber process, including state symbols.

2.2 Conditions

State the typical temperature, pressure and catalyst used.

Condition	Typical value / material
Temperature	
Pressure	
Catalyst	

3. Temperature compromise

Temperature change	Effect on NH_3 equilibrium yield	Effect on rate	Industrial consequence
Lower temperature			
Higher temperature			

3.1 Explain

Explain why industry does not simply use the lowest possible temperature.

3.2 Compromise

Explain why about 450 °C represents a compromise.

4. Pressure compromise

4.1 Mole count

For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, count the gas moles on each side.

4.2 Equilibrium

Explain why increasing pressure increases the equilibrium yield of ammonia.

4.3 Economics

Explain why industry does not simply use the highest possible pressure.

4.4 Compromise

Explain why about 200 atm can be described as an industrial compromise.

5. Catalyst

5.1 Role

Explain the role of the iron catalyst in the Haber process.

5.2 Equilibrium position

Explain why the catalyst does not increase the equilibrium yield at a fixed temperature and pressure.

6. Haber Process Compromise

Option	Equilibrium yield	Reaction rate	Economic / engineering consideration
Low temperature			
High temperature			
Low pressure			
Very high pressure			
Iron catalyst			

6.1 Overall judgement

Summarise why industrial conditions must balance yield, rate and economics.

7. Haber Process Decision Challenge

Plant X can operate at 350, 450 or 600 °C and at 100, 200 or 400 atm. An iron catalyst is available.

7.1 Recommendation

Recommend a realistic combination and justify it using equilibrium yield, reaction rate and economic/engineering effects.

8. Error analysis

8.1

Correct the statement: Use the lowest possible temperature because it gives the highest ammonia yield.

8.2

Correct the statement: Use the highest possible pressure because it gives the highest ammonia yield.

8.3

Correct the statement: The iron catalyst increases the equilibrium yield of ammonia.

8.4

Correct the statement: Higher temperature always improves industrial ammonia production.

9. Exam style questions

9.1 [3]

State typical temperature, pressure and catalyst used in the Haber process.

9.2 [3]

Explain why a lower temperature is not used even though it would increase equilibrium yield.

9.3 [2]

Explain why increasing pressure increases ammonia yield.

9.4 [3]

Explain why industrial pressure is a compromise.

9.5 [2]

Explain the role of the iron catalyst.

10. Final knowledge check

10.1

Write the Haber equation.

10.2

State the three typical operating conditions.

10.3

Why does lower temperature favour ammonia?

10.4

Why is a very low temperature not used?

10.5

Why does high pressure favour ammonia?

10.6

Why is extremely high pressure not used?

10.7

What does the catalyst change and what does it not change?